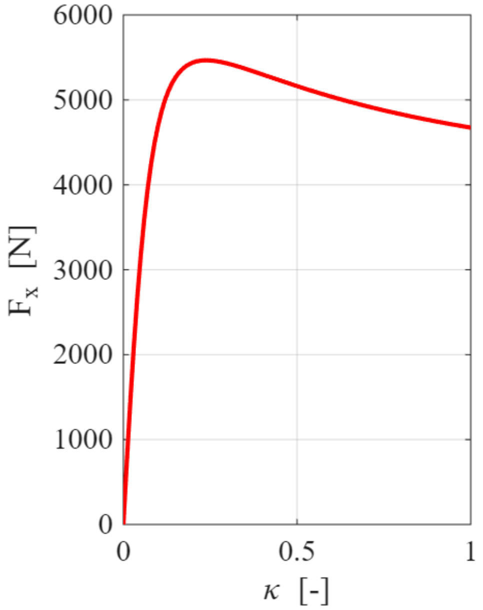
VEHICLE DYNAMICS AND CONTROL- TIRE LAB

CAR SEGMENT - Sports Car

This report contains the results for the Vehicle Dynamics and Control Lab on Tires. The car segment chosen for the simulations is Sports Car.

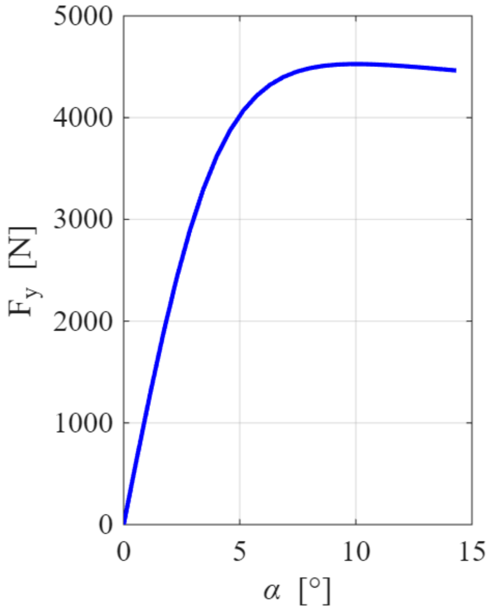
Steady State Analysis

a. **Fx vs κ in steady-state conditions Fz = Fz0, α = 0, μ = 1**



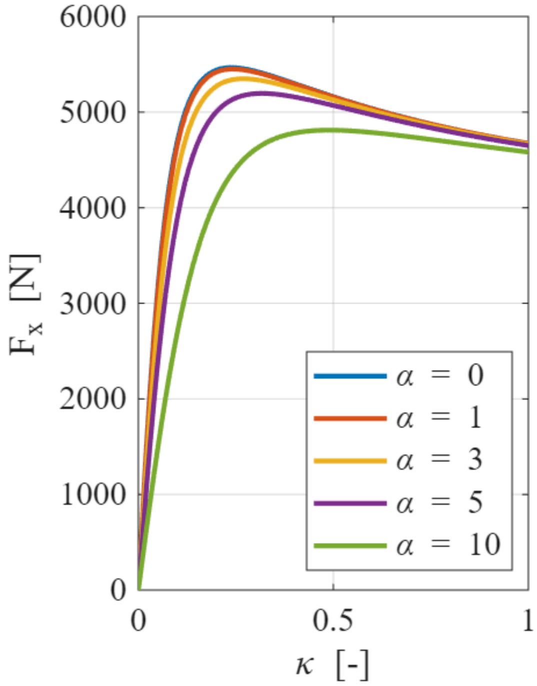
From the plot, we can find the peak force to be 5465N at a slip ratio of 0.23°

b. **Fy vs α in steady-state conditions Fz = Fz0, κ = 0, μ = 1**



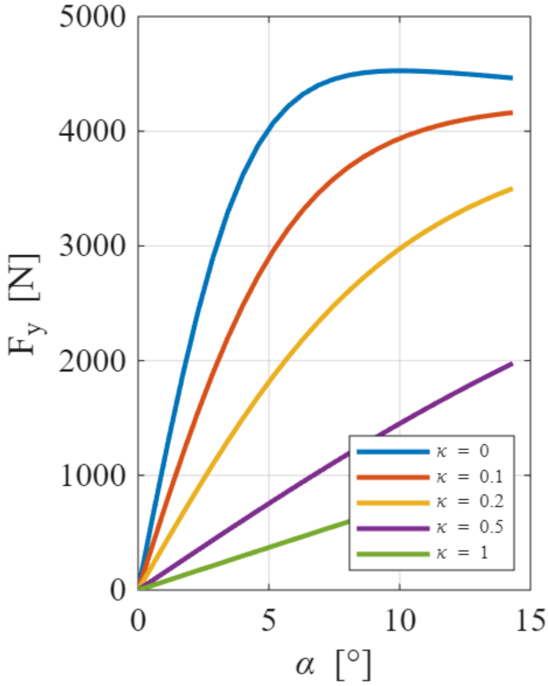
From the plot, we see that the peak lateral force is 4526N at a slip angle of 9.74°

c. Fx vs κ in steady-state conditions Fz = Fz0, μ = 1, for different slip angles



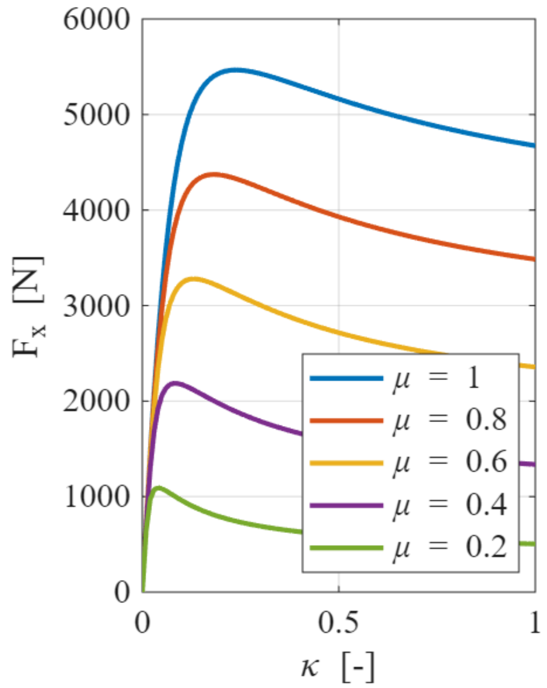
From this plot, we can conclude that the Longitudinal Force that is generated decreases with increasing slip angles for the same slip ratio.

d. **Fy vs α in steady-state conditions Fz = Fz0, μ = 1, for different slips**



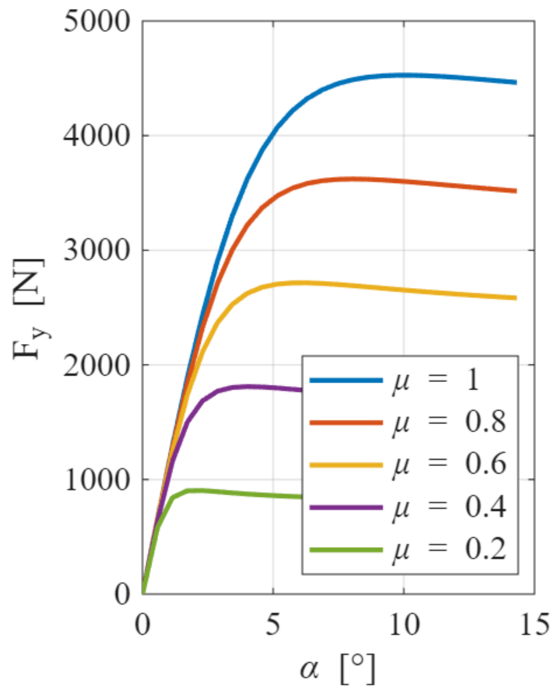
In this plot, it is visible that the lateral force generated at the tire decreases for increasing slip for the same slip angles. However, it takes larger slip angles to generate the peak lateral force.

e. Fx vs κ in steady-state conditions Fz = Fz0, α = 0, for different friction coefficients



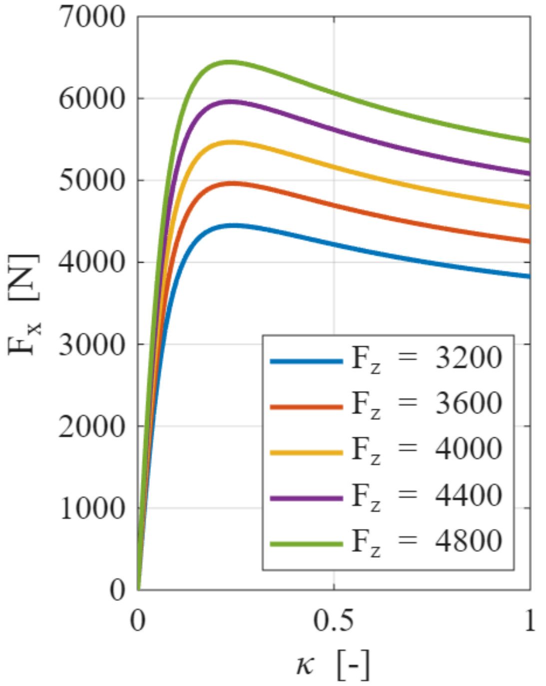
Here, it can be seen that the longitudinal force generated is increasing for the same slip with increasing value of friction coefficient.

f. Fy vs α in steady-state conditions Fz = Fz0, κ = 0, for different friction coefficients



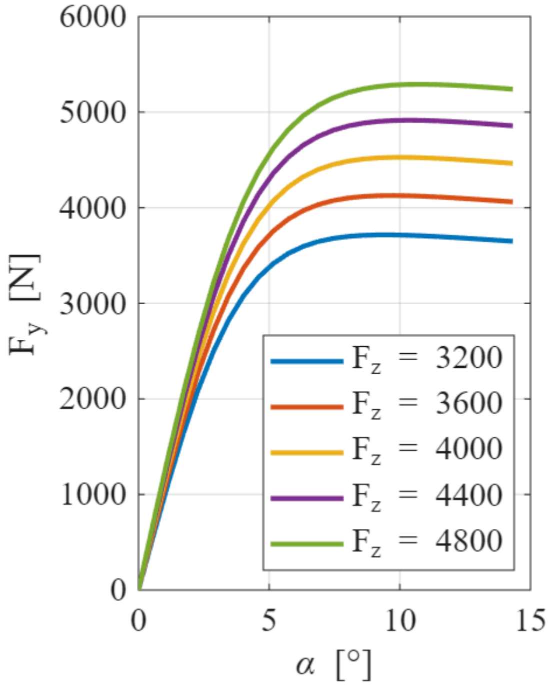
Similarly, in case of lateral force, the force generated increases for increasing values of friction coefficients at the same slip angle. It can also be seen that the peak lateral force is generated for higher slip angles for bigger values of μ.

g. Fx vs κ in steady-state conditions, α = 0, μ = 1, for different normal loads



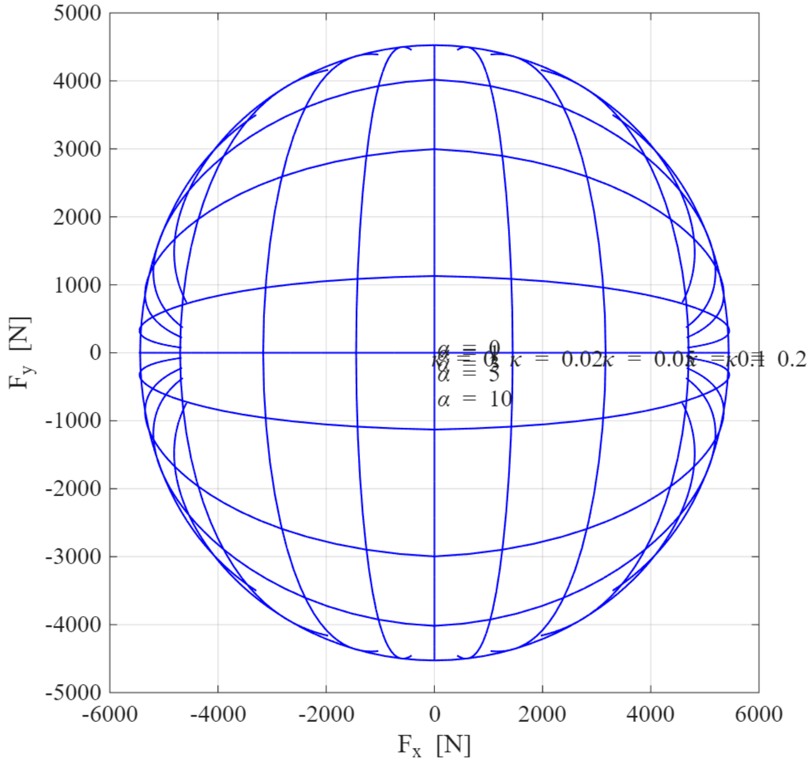
The following curve depicts the Tire Load Sensitivity for longitudinal force.
The peak longitudinal force generated increases with increasing normal load, Fz, on tires.

h. Fy vs α in steady-state conditions, κ = 0, μ = 1, for different normal loads



The tire load sensitivity to lateral forces is shown. Increasing Fz leads to higher lateral forces which are generated at almost the same slip angles.

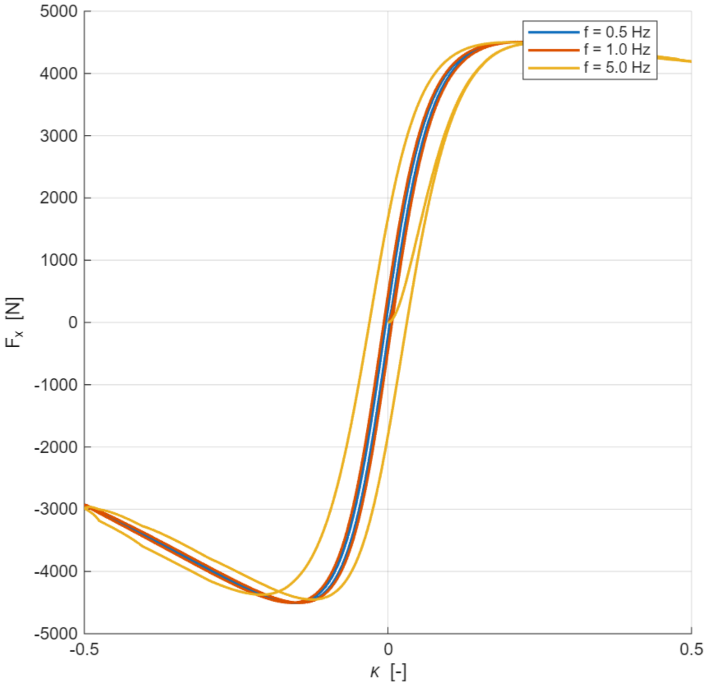
i. Fx vs Fy in steady-state conditions Fz = Fz0, μ = 1



The effect of combined lateral and longitudinal forces is shown for a friction coefficient of 1.

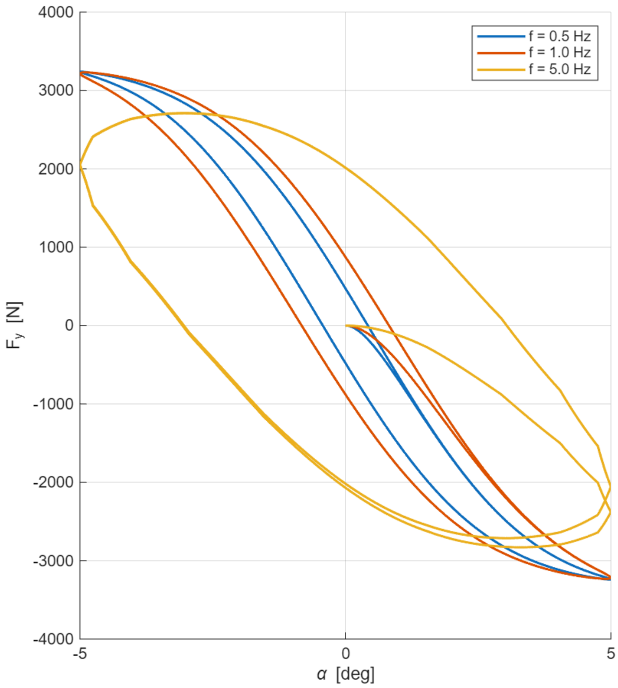
TRANSIENT ANALYSIS

a. Fx vs κ in transient conditions Fz = Fz0, α = 0, μ = 1, for different frequencies



The longitudinal force–slip ratio relationship exhibits a clear frequency-dependent hysteresis under transient conditions. As excitation frequency increases, the hysteresis loop widens and shifts, indicating greater relaxation effects and a delayed tire force response, while the peak longitudinal force remains nearly unchanged.

b. Fy vs α in transient conditions Fz = Fz0, κ = 0, μ = 1, for different frequencies



Similarly for lateral forces, higher excitation frequencies produce larger lateral force–slip angle hysteresis loops, reflecting increased tire relaxation effects and slower cornering response.